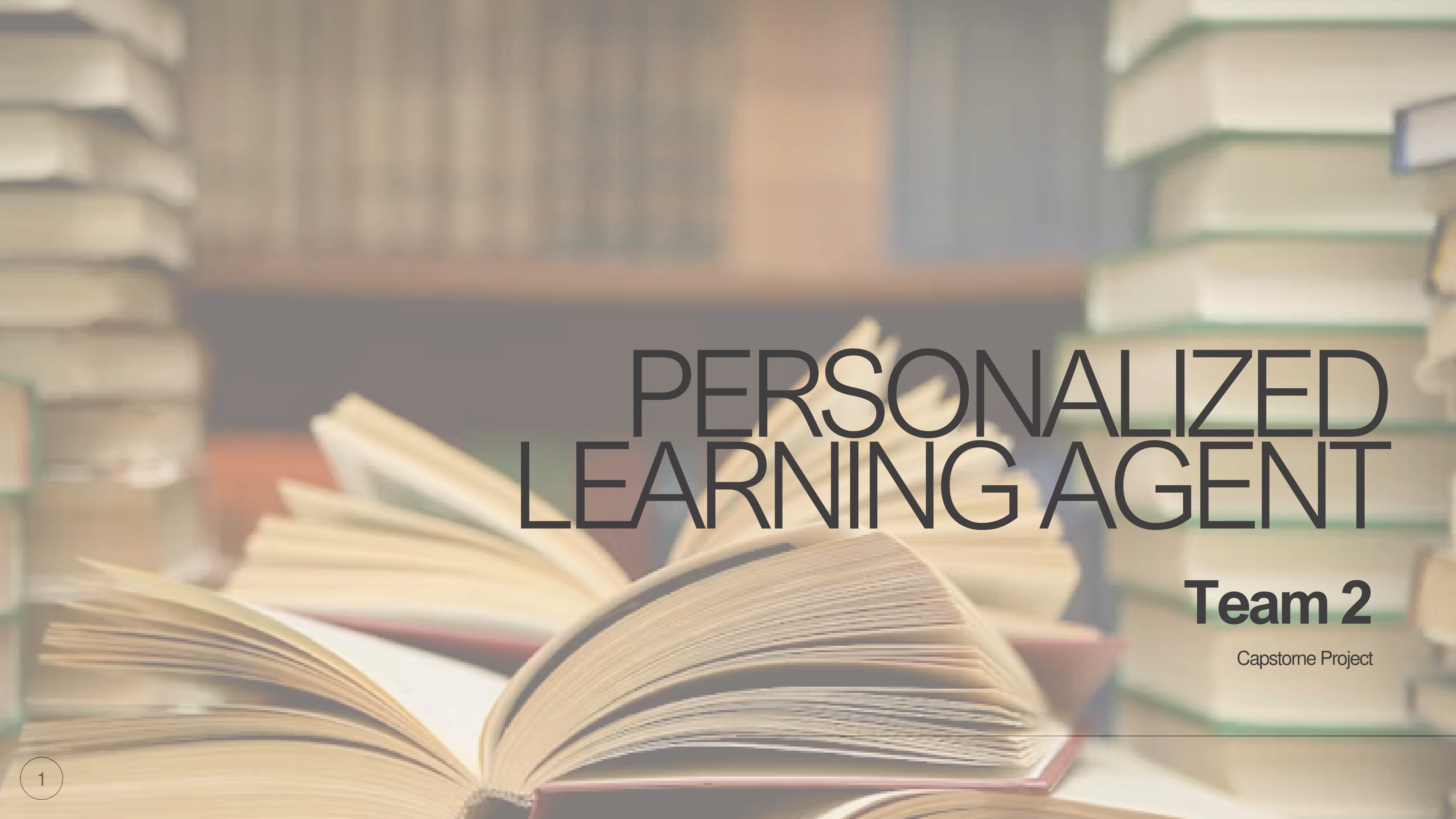




PERSONALIZED LEARNING AGENT

Team 2

Capstone Project

PERSONALIZED LEARNING AGENT

AIM

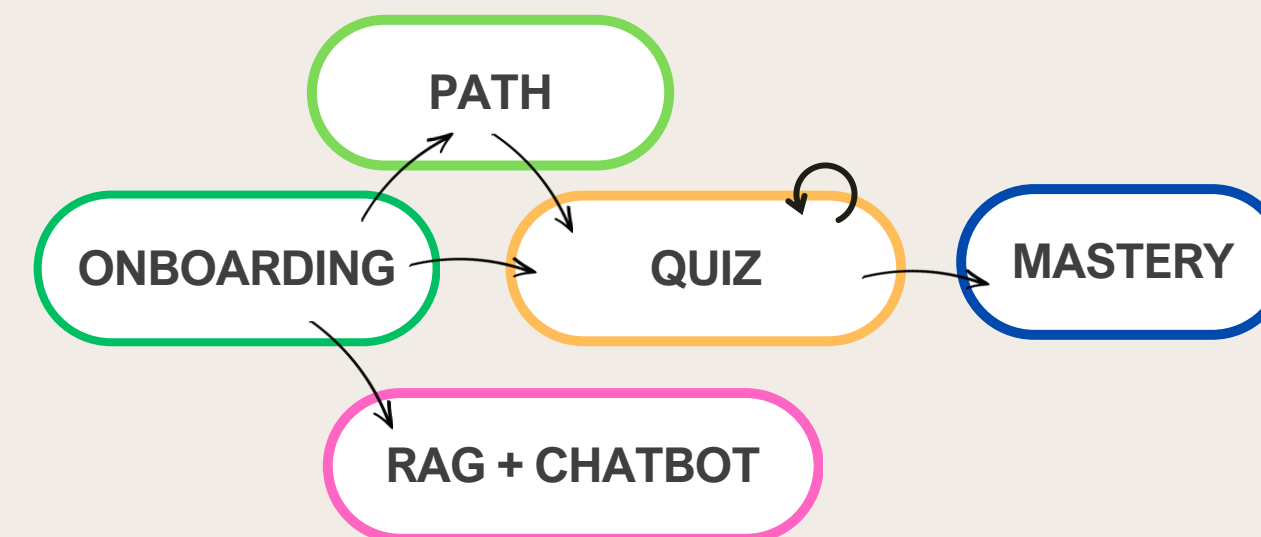
- Our goal for this project
- There are several products online, that provide... such as ...which allows...
- However, we noticed....

AI APPROACH

1. Gen AI (Llama-3.3-70b language model)
 - a. Diagnostic question generation (difficulty plan decided before calling the LLM)
 - b. Adaptive quiz generation — two separate LLM calls: one to write questions, one to classify each question's Bloom level + concept count + reasoning steps
 - c. Learning path roadmap generation — LLM generates step nodes with prerequisite IDs, then a second call generates 4 preview terms per step
 - d. Quick Study RAG answers — strict prompt rules: cite sources, walk through examples step-by-step, return JSON with confidence score
2. RAG is implemented as a hybrid retrieval system (not plain cosine similarity):
 - a. $\text{score} = 0.7 \times \text{semantic_similarity} + 0.3 \times \text{keyword_overlap}$
 - b. Adds neighbour chunk expansion (pulls chunks ± 1 adjacent) to avoid cutting textbook examples mid-flow.

SOLUTION OVERVIEW

- Full Architecture Flow:



ADAPTIVE QUIZ EFFECTIVENESS

DIFFICULTY CALIBRATION IMPLEMENTATION

LLM GENERATED
QUESTION

LLM FEATURE
EXTRACTION

DIFFICULTY SCORE

DIFFICULTY LABEL

Educational Features

- Bloom Level
- Concept Count
- Reasoning Complexity

EVALUATION PIPELINE

ARC EASY & ARC
CHALLENGE

LLM FEATURE
EXTRACTION

DIFFICULTY SCORE
COMPUTATION

ARC LABELS

EVALUATION METRICS & RESULTS

- Accuracy
- ROC-AUC
- Mean Difficulty Gap

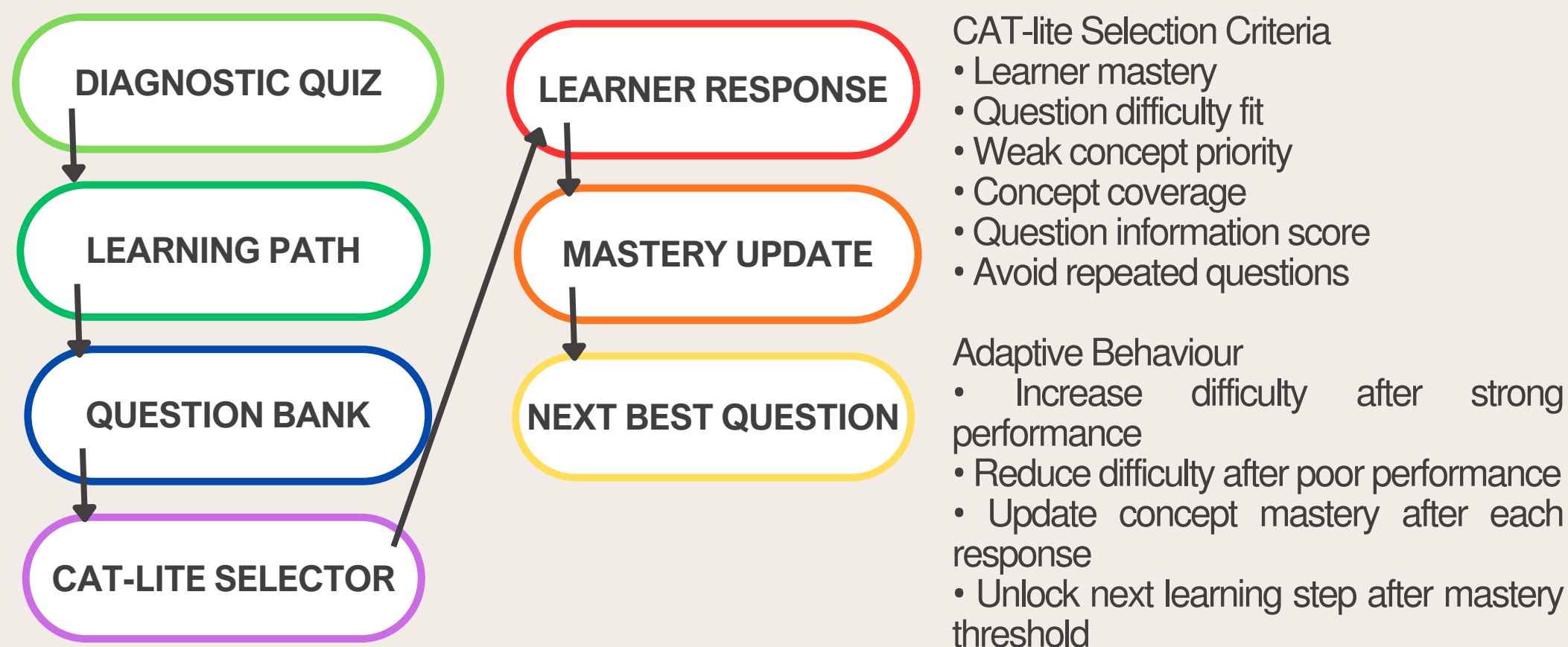
Latest Results
Accuracy: 0.53
ROC-AUC: 0.54
Mean Difficulty Gap: 0.131

Key Findings

- Challenge questions received higher average scores.
- ARC is used as a proxy benchmark for educational difficulty.

ADAPTIVE QUIZ EFFECTIVENESS

CAT-LITE IMPLEMENTATION



CAT-LITE EVALUATION USING SYNTHETIC LEARNERS

Simulation Setup

- 5 learner personas
- 30 runs per persona
- 15-question adaptive quiz
- 72-question synthetic bank
- Probability model with ability, guess & slip
- Metrics collected each question

Key Results

Beginner difficulty: 1.63
 Intermediate difficulty: 2.63
 Advanced difficulty: 3.56

Findings

- Difficulty matched learner ability
- Low average difficulty mismatch (~0.12–0.13)
- Full concept coverage with targeted revisits
- Mastery updates were conservative
- Pass threshold (0.8) proved difficult to reach

MASTERY IMPLEMENTATION

BKT estimates the probability that the learner knows a concept, instead of only showing correct divided by answered.

EARLIER

- Correct answer increased mastery by a fixed amount.
- Wrong answer decreased mastery by a fixed amount.
- Could not model guessing or careless mistakes.

BKT APPROACH

- Tracks mastery per concept.
- Correct/wrong answer becomes probabilistic evidence.
- Models prior knowledge, learning, guessing, and slipping.
- Better explanation for adaptive learning.

HOW CONCEPT MASTERY UPDATES

These defaults are conservative because we do not yet have enough real learner data for calibration.

PRIOR = 0.25

- Starting belief the learner already knows the concept.

TRANSIT = 0.12

- Chance the learner learns after one practice opportunity.

GUESS = 0.2

- Chance of a correct answer without knowing the concept.

SLIP = 0.10

- Chance of a wrong answer despite knowing the concept.

EXAMPLE: ONE CORRECT ANSWER CAN MOVE A NEW CONCEPT FROM 25.0% TO ABOUT 64.8%. ONE WRONG ANSWER CAN MOVE IT TO ABOUT 15.5%.

MASTERY IMPLEMENTATION

WHAT THE UI SHOWS

The dashboard separates BKT mastery from raw accuracy so the result is explainable.

RAW ACCURACY

- **Formula:** correct / answered.
- **Example:** 3 correct out of 4 = 75%.
- **Useful as evidence, but not a true mastery estimate.**

BKT MASTERY

- Reads `concept_mastery.mastery_probability`.
- **Averages concept probabilities per preview term.**
- **Example:** Data Structures Definition = 52.5% BKT, 75.0% accuracy.

HOW WE VALIDATE THE SYSTEM

We evaluate both the adaptive quiz behavior and the mastery estimate quality.

ADAPTIVE QUIZ EVALUATION

- Synthetic learner personas.
- 15-question adaptive quiz runs.
- Difficulty mismatch and concept coverage.
- Selector should target ability and weak concepts.

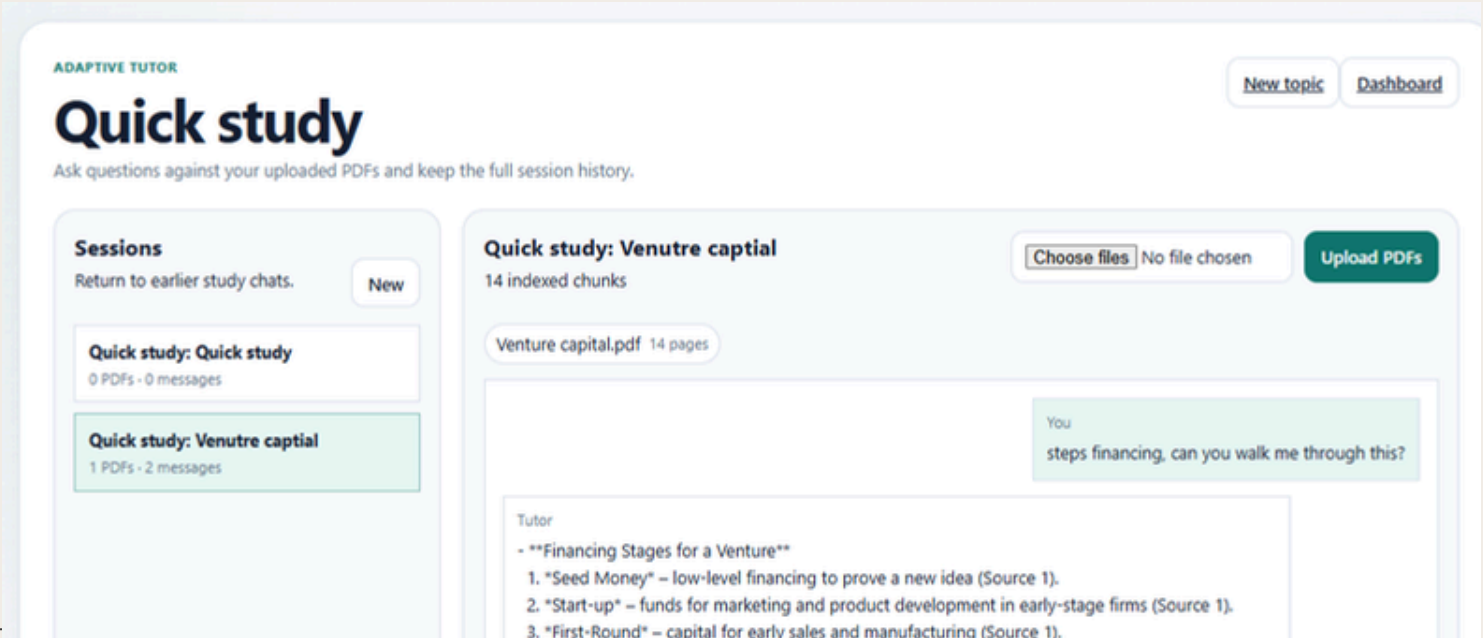
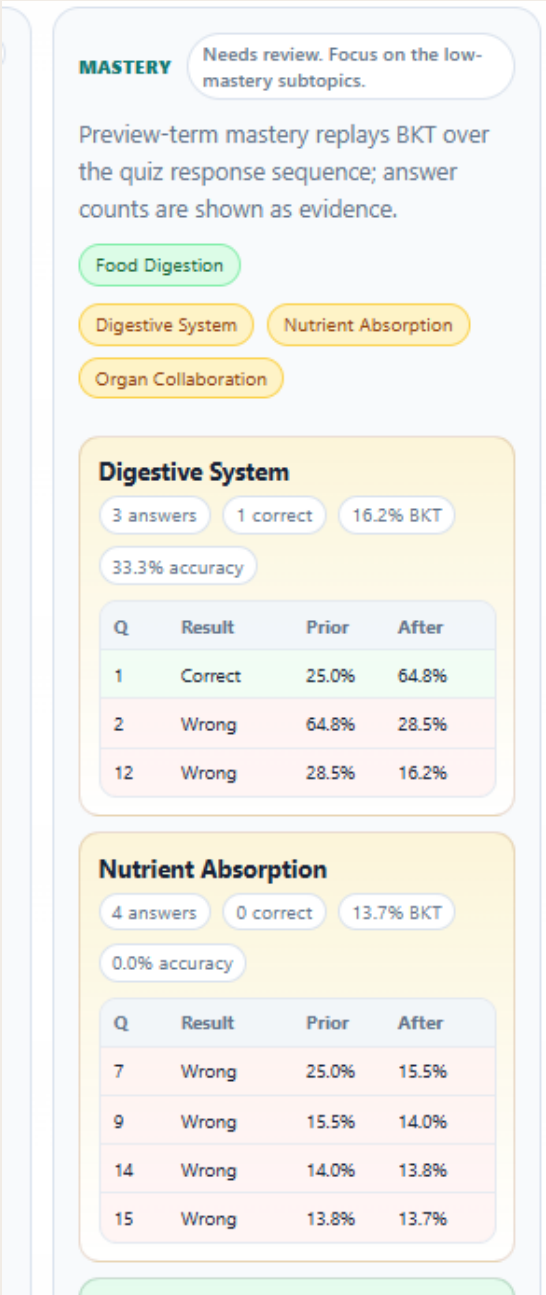
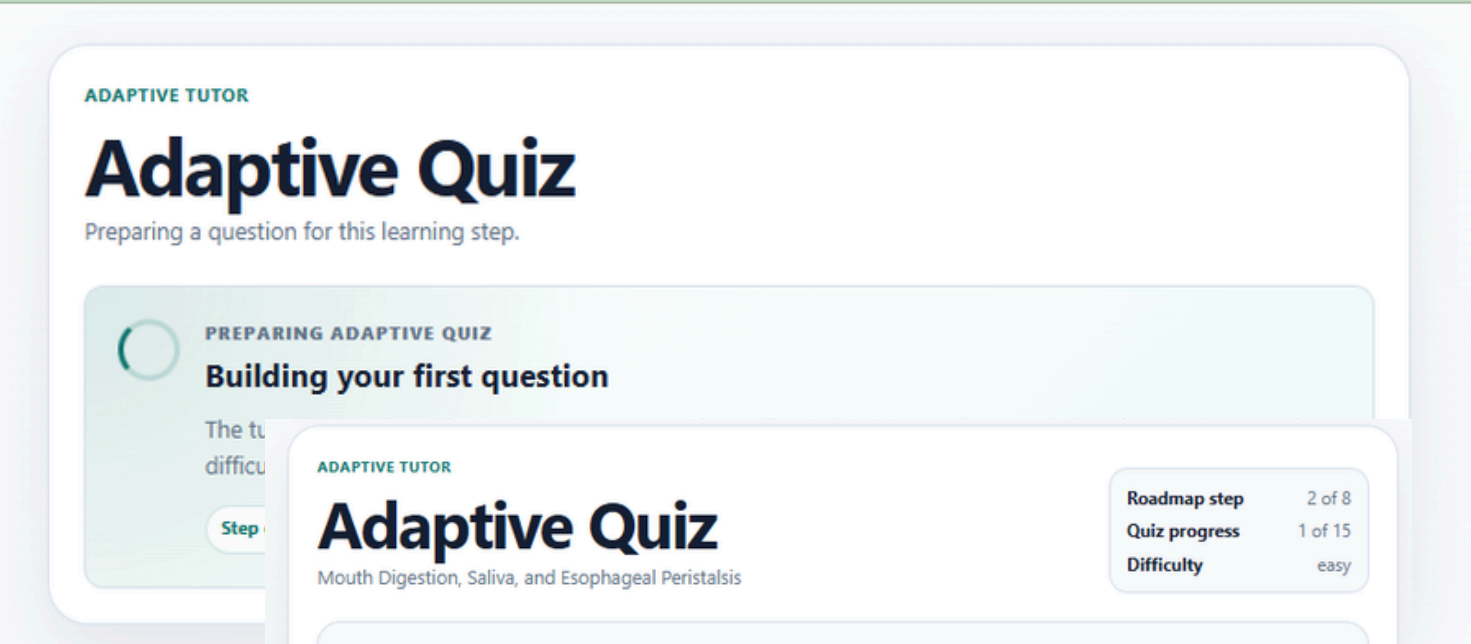
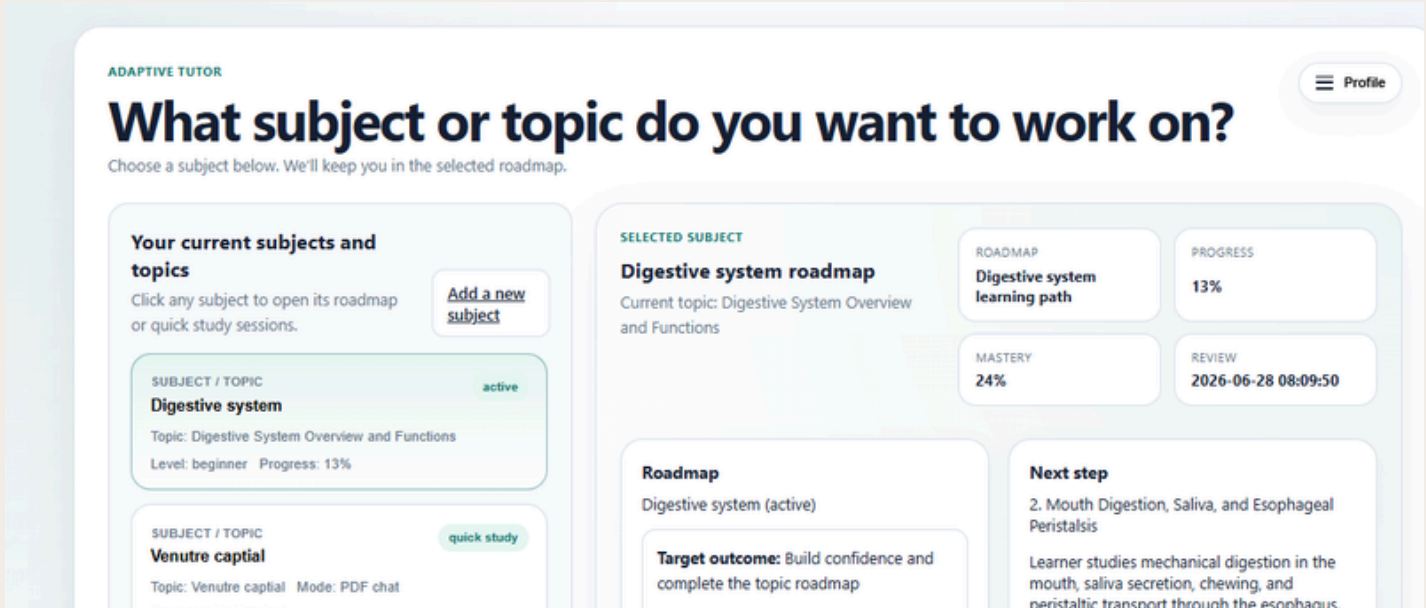
DIFFICULTY CALIBRATION

- Compare computed difficulty with ARC labels.
- Track accuracy, ROC-AUC, difficulty gap.
- Feature-based scoring remains explainable.

BKT MASTER QUALITY

- Correct answers should increase mastery.
- Wrong answers should decrease mastery.
- Repeated success approaches mastery gradually.
- Concepts not assessed should not change.

DEMO



A SIMPLE END-TO-END STORY FOR THE PRESENTATION DEMO.

